

Network and Clustering Approaches to Portfolio Diversification on FTSE 100

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We construct diversified portfolios for FTSE 100 equities using correlation based methods. From the training period correlation matrix we build a sparse network and apply graph theoretic selection rules including degeneracy ordering and inverse eigenvector centrality. We complement these network portfolios with clustering approaches using hierarchical methods and Hierarchical Equal Risk Contribution. We also include a K-means portfolio based on historical return and volatility rather than correlation. All portfolios were tested from 2019 to 2025 and achieved greater performance compared to FTSE 100 index.

I. DATA CONFIGURATION

We evaluate these strategies using an out of sample backtest where we define a training period from 2016 to 2018 to calibrate the models and calculate portfolio weights, and then validate performance over a separate test window from 2019 to 2025. To test the effectiveness of our diversification logic we use a buy and hold strategy and do not rebalance during the test phase, which allows us to observe how the diversification performs when our initial exposures remain constant. Our dataset contains daily closing prices sourced from Bloomberg Terminal and all returns are annualized assuming 252 trading days.

II. DATA PROCESSING AND CORRELATION STRUCTURE

We denote the closing price of stock i on day t as $P_{i,t}$. The daily log returns are calculated as

$$r_{i,t} = \log(P_{i,t}) - \log(P_{i,t-1}). \quad (1)$$

We use the data from the training window to compute the sample Pearson correlation matrix ρ and the covariance matrix Σ which serve as the primary inputs for the network and hierarchical clustering models.

While we use log returns to estimate the correlation structure we use simple arithmetic returns to calculate the final portfolio performance ensuring the results match the actual profit and loss of the buy and hold strategy.

Figure 1 displays the correlation matrix derived from the training period. The heatmap shows that correlations are not uniform across the index and we observe distinct blocks of high correlation which generally indicate stocks with shared sector exposures.

III. PORTFOLIO CONSTRUCTION

We calculate portfolio weights w_i such that $w_i \geq 0$ and $\sum_i w_i = 1$ where t_0 denotes the start of the test period.

Stock Correlation Heatmap (2016-01-01 to 2018-12-31)

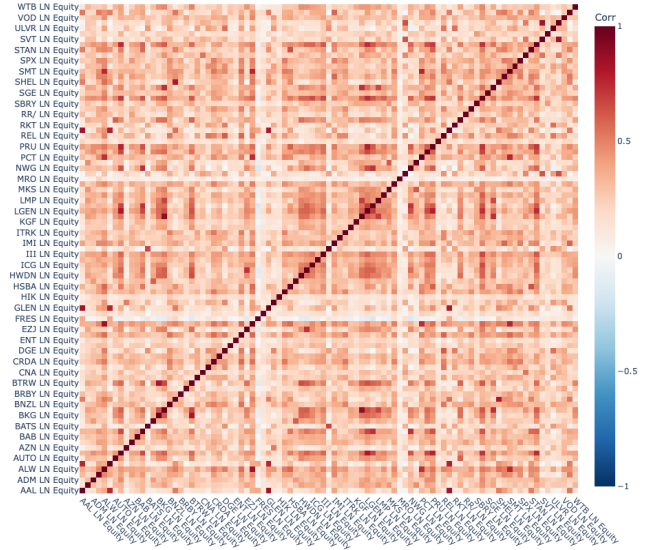


FIG. 1. Training period correlation heatmap.

The portfolio value V_t is given by

$$V_t = \sum_i w_i \frac{P_{i,t}}{P_{i,t_0}}. \quad (2)$$

The daily portfolio return is then calculated as

$$R_t^{(p)} = \frac{V_t}{V_{t-1}} - 1. \quad (3)$$

This return series tracks the buy and hold performance from the initial allocation at t_0 .

IV. NETWORK BASED ALLOCATIONS

A. Building the Correlation Network

The market is represented as a graph $G = (V, E)$ with one node per stock [1]. Using training period correlations we keep only strong positive links to obtain a sparse graph. Let ρ_{ij} be the correlation between stocks i and j .

Degeneracy Network (yellow = selected stocks)

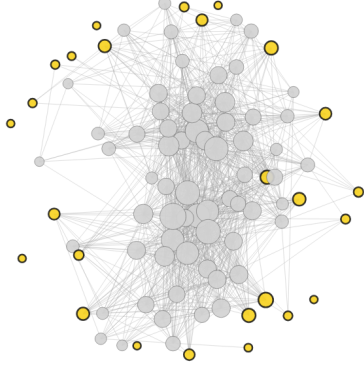


FIG. 2. Thresholded correlation network used for the degeneracy selection rule.

We add an undirected edge $(i, j) \in E$ when $\rho_{ij} > \tau$ and set the edge weight to $w_{ij} = \rho_{ij}$.

The threshold τ is set at the 75th percentile of the off diagonal correlations ensuring the network emphasizes the strongest relationships while controlling sparsity [2].

B. Degeneracy Selection Portfolio

This rule selects a subset of stocks that are weakly connected in the thresholded network [1]. The selection is based on graph degeneracy using the k core decomposition where we prioritize less embedded nodes so that the chosen set has limited overlap in strong correlation links [3]. If two stocks are linked by a strong correlation edge then holding both adds little diversification under the network definition.

The procedure has two steps. First we include all isolated nodes with degree zero. Second we apply a greedy selection on the remaining subgraph that avoids adjacency adding a node only if it is not adjacent to any already selected node. Nodes are processed in order of increasing core number favoring less embedded nodes.

Equal weights are assigned to the selected set S as

$$w_i = \begin{cases} \frac{1}{|S|} & \text{if } i \in S, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

C. Inverse Eigenvector Centrality Portfolio

Eigenvector centrality measures how connected a node is to other well connected nodes [1]. Highly central stocks tend to lie in the core of the dependence structure. To diversify we use inverse centrality assigning greater weight to less central stocks and reducing exposure to the core while remaining fully invested [4].

Eigenvector Centrality Network

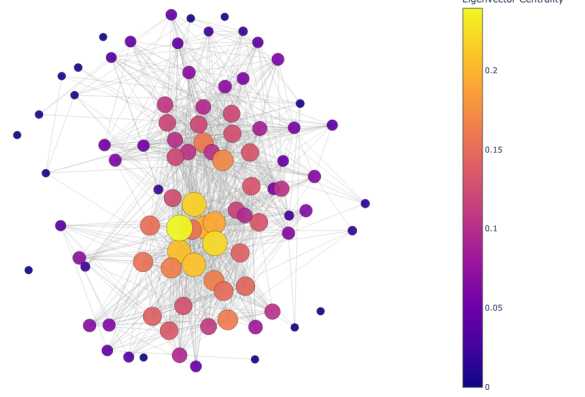


FIG. 3. Thresholded network colored by eigenvector centrality.

Let $\mathbf{A}$ be the weighted adjacency matrix of the thresholded network. Eigenvector centrality satisfies

$$\lambda \mathbf{s} = \mathbf{A} \mathbf{s}. \quad (5)$$

We transform centrality into long only weights by tilting against the most central nodes as

$$\tilde{s}_i = \left(\max_j s_j - s_i \right) + \varepsilon, \quad w_i = \frac{\tilde{s}_i}{\sum_j \tilde{s}_j}, \quad (6)$$

with $\varepsilon = 10^{-12}$ to avoid zero weights.

V. CLUSTERING BASED ALLOCATIONS

A. Cluster Equal Weight Portfolio

We employ hierarchical clustering to partition the asset universe using the correlation based distance metric

$$d_{ij} = \sqrt{\frac{1 - \rho_{ij}}{2}}. \quad (7)$$

A dendrogram is constructed using average linkage clustering and cut to obtain up to 15 distinct groups with a minimum cluster size of 3 stocks. Let C_k denote the set of stocks in cluster k and $|C_k|$ be the number of stocks in that cluster. We assign weights to ensure equal capital allocation across clusters and equal allocation to assets within each cluster as

$$w_i = \frac{1}{K} \cdot \frac{1}{|C_{c(i)}|}, \quad (8)$$

where $c(i)$ is the cluster index for stock i and K is the final number of clusters.

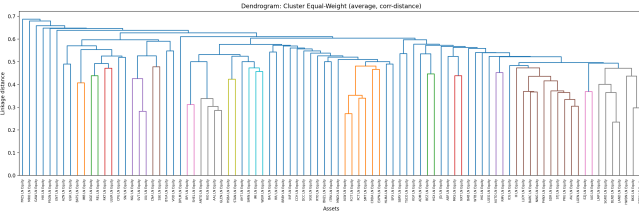


FIG. 4. Dendrogram used for hierarchical clustering.

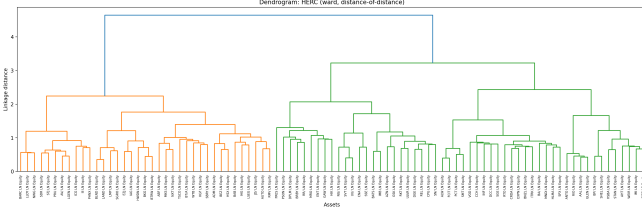


FIG. 5. Dendrogram used by the hierarchical risk allocation method.

B. Hierarchical Equal Risk Contribution

While the cluster equal weight method assigns uniform importance to distinct groups HERC allocates capital based on the estimated risk of each hierarchical branch [5].

Using the training covariance matrix Σ we compute the pairwise correlations ρ_{ij} and standard deviations $\sigma_i = \sqrt{\Sigma_{ii}}$. We calculate the base distance d_{ij} and subsequently the second order distance metric to capture similarities in correlation profiles as

$$\tilde{d}_{ij} = \|D_{\cdot i} - D_{\cdot j}\|_2, \quad (9)$$

where D is the matrix of base distances $\{d_{ij}\}$. We apply Ward linkage to $\tilde{d}_{ij}$ to build the hierarchy.

Portfolio weights are derived by recursively bisecting the cluster tree into subsets L (left) and R (right). At each bifurcation the variance of a subset is estimated using the inverse variance weighted portfolio of its constituents as

$$\text{Var}(L) = \mathbf{v}_L^\top \Sigma_L \mathbf{v}_L, \quad \mathbf{v}_L = \frac{\text{diag}(\Sigma_L)^{-1}}{\mathbf{1}^\top \text{diag}(\Sigma_L)^{-1}}, \quad (10)$$

and similarly for R . Capital is then allocated between the two branches inversely to their variance as

$$\alpha_L = \frac{\text{Var}(R)}{\text{Var}(L) + \text{Var}(R)}, \quad \alpha_R = \frac{\text{Var}(L)}{\text{Var}(L) + \text{Var}(R)}. \quad (11)$$

This top down risk allocation proceeds until the leaf nodes (individual assets) are reached.

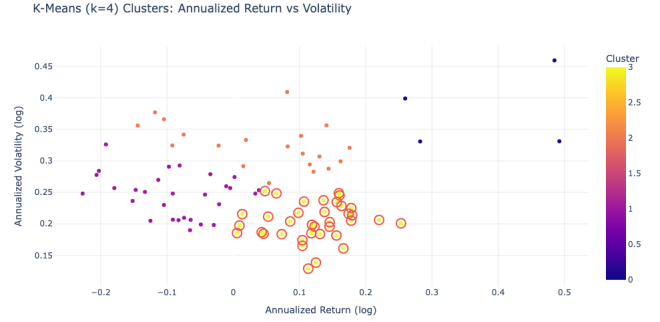


FIG. 6. K-means clustering in the risk return space.

C. K-Means Return and Volatility Portfolio

This strategy applies K-means clustering to group assets based on their historical risk return profiles. For each stock i we compute the annualized mean log return and volatility over the training period as

$$\hat{\mu}_i = 252 \cdot \bar{r}_i, \quad \hat{\sigma}_i = \sqrt{252} \cdot \text{sd}(r_i). \quad (12)$$

We map these points $(\hat{\mu}_i, \hat{\sigma}_i)$ into the risk return space and partition them into $K = 4$ clusters. We then rank each cluster c by its aggregate reward to risk ratio as

$$\text{Score}(c) = \frac{\bar{\hat{\mu}}_c}{\hat{\sigma}_c}. \quad (13)$$

We select the top cluster subject to a minimum size of 7 stocks. Capital is allocated equally to the constituents within the selected cluster.

VI. PERFORMANCE MEASUREMENT

The annualized mean return and volatility are defined as

$$\hat{\mu} = 252 \cdot \overline{R^{(p)}}, \quad \hat{\sigma} = \sqrt{252} \cdot \text{sd}(R^{(p)}). \quad (14)$$

We report the Sharpe ratio assuming a zero risk free rate as

$$\text{Sharpe} = \frac{\hat{\mu}}{\hat{\sigma}}. \quad (15)$$

To measure diversification efficiency we calculate the Diversification Ratio as

$$\text{DR} = \frac{\mathbf{w}^\top \boldsymbol{\sigma}}{\sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}}, \quad (16)$$

where $\mathbf{w}$ represents the portfolio weights and $\boldsymbol{\sigma}$ and Σ are computed from the test period returns.

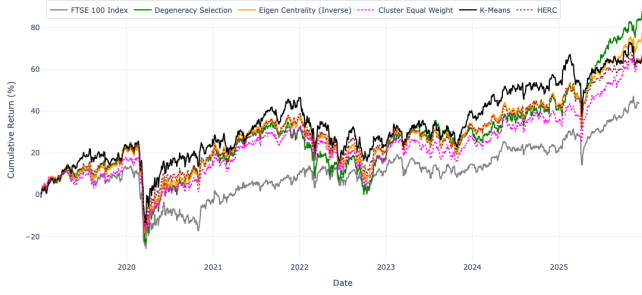


FIG. 7. Cumulative returns over the test window.

TABLE I. Test window performance metrics.

Portfolio	Mean (%)	Vol (%)	Sharpe	DR
FTSE 100	6.50	16.05	0.40	—
Degeneracy	10.20	16.88	0.60	1.93
Inv. eigen.	9.31	15.99	0.58	1.86
Cluster equal	8.31	15.13	0.54	1.88
K-means	8.37	16.07	0.52	1.68
HERC	8.35	15.35	0.54	1.81

VII. RESULTS

Fig. 7 plots the cumulative performance relative to the FTSE 100.

Table I details the annualized metrics. The degeneracy portfolio achieves a Sharpe ratio of 0.604 with return of 10.20%. The inverse eigenvector centrality portfolio achieves a Sharpe ratio of 0.583 with return of 9.32%. The cluster equal portfolio achieves a Sharpe ratio of 0.549 with return of 8.31% and volatility of 15.13%. HERC achieves a Sharpe ratio of 0.544 with return of 8.36% and volatility of 15.35%. The K-means portfolio achieves a Sharpe ratio of 0.521 with return of 8.38%. All strategies outperform the FTSE 100 benchmark which posted a Sharpe ratio of 0.405.

The Diversification Ratios range from 1.68 to 1.94. The correlation based methods achieve diversification ratios between 1.82 and 1.94 while the K-means approach achieves 1.68.

The degeneracy portfolio selected weakly connected stocks from the network periphery. The inverse eigenvector centrality portfolio downweighted stocks in the highly correlated market core. The cluster equal portfolio spread capital equally across correlation clusters. HERC allocated capital based on the hierarchical risk structure. The K-means portfolio selected stocks based on historical returns and volatility.

VIII. CONCLUSION

The buy and hold approach over a seven year test period provides an assessment of diversification quality. The strategies use simple allocation rules without optimization of portfolio weights or rebalancing mechanisms. Transaction costs and position size constraints would also affect realized performance.

Future work could combine network based selection with mean variance optimization or alternative risk parity methods or even networks that encode tail or rank dependence in addition to linear correlation [6] could also improve diversification.

Appendix A: Portfolio Weights

Table II reports portfolio allocation for the selected stocks across the five strategies. The table shows only stocks receiving non-zero allocations under at least one strategy.

Appendix B: Reproducibility

The implementation and instructions to reproduce the results are available at

<https://github.com/AUEBs-INVESTMENT-FINANCE-CLUB/network-clustering-portfolios>

- [1] F. J. Fabozzi, M. Focardi, and C. Jonas, *Network Models in Finance: Expanding the Tools for Portfolio and Risk Management*, Wiley Finance (2016).
- [2] M. Tumminello, T. Aste, T. Di Matteo, and R. N. Mantegna, “A tool for filtering information in complex systems,” *Proc. Natl. Acad. Sci. USA* **102**, 10421–10426 (2005).
- [3] F. Pozzi, T. Di Matteo, and T. Aste, “Spread of risk across

financial markets: better to invest in the peripheries,” *Sci. Rep.* **3**, 1665 (2013).

- [4] G. Peralta and A. Zareei, “A network approach to portfolio selection,” SSRN Working Paper (2016).
- [5] T. Raffinot, “Hierarchical clustering-based asset allocation,” *J. Portfolio Manag.* **44**(2), 89–99 (2018).
- [6] M. Jaeger and D. Marinelli, “Network diversification for a robust portfolio allocation,” SSRN 4068889 (2022).

TABLE II. Portfolio weights by ticker (%).

Ticker	Degeneracy (%)	Inv. eigen. (%)	Cluster eq (%)	K-means (%)	HERC (%)
ABF	3.846	1.233	0.490	0.000	0.616
ADM	0.000	1.235	0.490	2.778	1.314
ALW	0.000	0.565	0.725	2.778	3.106
AUTO	3.846	0.966	0.490	0.000	0.676
AZN	3.846	1.661	1.667	2.778	1.669
BAB	3.846	1.109	0.490	0.000	0.878
BATS	3.846	1.666	1.667	0.000	1.152
BEZ	0.000	1.192	0.490	2.778	1.243
BNZL	0.000	1.189	0.725	2.778	2.282
BP	3.846	1.263	1.190	2.778	0.899
BRBY	3.846	1.528	0.725	0.000	0.977
BT/A	3.846	1.570	2.381	0.000	1.111
CCH	3.846	1.443	0.725	2.778	1.174
CNA	3.846	1.681	2.381	0.000	0.672
CPG	3.846	1.526	1.667	2.778	2.182
CRDA	0.000	0.960	0.725	2.778	1.448
DCC	0.000	1.068	0.725	2.778	1.810
DGE	0.000	1.395	1.667	2.778	1.636
DPLM	3.846	1.695	0.725	2.778	1.738
ENT	3.846	1.695	0.490	0.000	1.373
EXPN	0.000	0.829	0.725	2.778	1.486
FCIT	0.000	0.481	0.725	2.778	3.008
FRES	3.846	1.668	1.190	0.000	0.620
GAW	3.846	1.695	0.725	0.000	1.083
GSK	0.000	1.518	1.667	2.778	1.982
HIK	3.846	1.695	0.725	0.000	0.738
HLMA	0.000	0.968	0.725	2.778	1.280
HSBA	0.000	0.861	1.190	2.778	1.678
HSX	3.846	1.358	0.490	2.778	1.785
IHG	0.000	0.752	0.490	2.778	0.851
III	0.000	0.303	0.490	2.778	0.537
IMB	0.000	1.665	1.667	0.000	1.453
INF	3.846	1.410	0.725	2.778	1.708
ITRK	0.000	1.264	0.725	2.778	1.079
JD	3.846	1.219	0.490	0.000	0.559
KGF	3.846	1.420	5.556	0.000	1.011
LMP	0.000	0.778	0.490	2.778	1.369
LSEG	0.000	0.608	0.490	2.778	1.365
MNDI	0.000	0.830	0.725	2.778	1.204
MRO	3.846	1.695	0.725	0.000	0.098
NG	0.000	1.638	2.381	0.000	2.904
PCT	0.000	0.923	0.725	2.778	0.775
PSON	3.846	1.695	1.190	0.000	0.900
REL	0.000	1.306	1.667	2.778	1.558
RKT	0.000	1.566	1.667	2.778	1.729
RR	3.846	1.599	0.725	0.000	0.901
RTO	3.846	1.284	0.725	2.778	1.900
SGRO	0.000	0.669	0.490	2.778	0.895
SHEL	0.000	1.306	1.190	2.778	0.948
SMIN	0.000	0.759	1.190	2.778	1.065
SMT	0.000	0.831	0.725	2.778	0.715
SN	0.000	1.375	1.667	2.778	1.532
SPX	0.000	1.010	0.725	2.778	1.150
SSE	0.000	1.364	2.381	0.000	2.396
SVT	0.000	1.669	2.381	0.000	1.670
TSCO	3.846	1.686	5.556	0.000	1.172
ULVR	0.000	1.539	1.667	2.778	1.794
UU	3.846	1.679	2.381	0.000	1.149
VOD	0.000	1.228	2.381	0.000	1.880